

Markscheme

Mathematics: analysis and approaches

Practice question bank

73. (a)

The sum to infinity of a geometric sequence is four times its first term, where $u_1 \neq 0$.

Find the common ratio r .

$$\frac{u_1}{1-r} = 4u_1 \quad (M1)$$

$$\frac{1}{1-r} = 4 \quad (M1)$$

$$r = 0.75 \quad \text{A1}$$

[3 marks]